Assignment #4-(GCCF)

SECTION B (2*4=8)

Q1. (CO4) What is Desktop virtualization?

Ans- Desktop virtualization is the process of separating the desktop environment and associated application software from the physical client device used to access it. It allows users to access their desktops remotely from any device, providing flexibility and central management of resources.

Q2. Define Virtualization.

Ans- Virtualization is the process of creating a virtual (rather than actual) version of something, such as a server, storage device, network or operating system. It allows multiple virtual instances of these resources to run simultaneously on a single physical hardware platform. This enhances efficiency, flexibility, and scalability in computing environments.

Q3. (CO4) What is Server Virtualization?

Ans- Server virtualization is the process of dividing a physical server into multiple virtual servers, each running its own operating system (OS) and applications. This allows for better utilization of server resources, improved scalability, and easier management. Server virtualization helps organizations reduce hardware costs, streamline IT infrastructure, and increase flexibility in deploying and managing applications.

Q4. (CO4) Differentiate between VM and Load Balancer.

Ans- Virtual Machine (VM) and Load Balancer are two different components used in modern computing environments, each serving distinct purposes:

1. Virtual Machine (VM):

- A VM is an emulation of a physical computer system, created through virtualization technology.
- It runs its own operating system (OS) and applications, isolated from other VMs on the same physical server.
- VMs enable the consolidation of multiple virtual servers onto a single physical server, optimizing resource utilization and improving scalability.

- They provide flexibility in deploying and managing software, enabling easier migration, replication, and backup of entire server environments.

2. Load Balancer:

- A Load Balancer is a network device or software component that distributes incoming network traffic across multiple servers (or VMs) to ensure optimal resource utilization, maximize throughput, and minimize response time.

- It helps prevent overload on individual servers by evenly distributing incoming requests among a pool of servers.

- Load balancers can perform various types of load distribution algorithms, such as round-robin, least connections, or weighted distribution, depending on the configuration and requirements.

- They enhance fault tolerance and high availability by redirecting traffic away from failed or overloaded servers to healthy ones, ensuring uninterrupted service for users.

Feature	VM	Load Balancer
Function	Runs applications and services	Distributes traffic among servers
Purpose	Provides isolated computing environment	Improves performance, reliability, scalability
Scalability	Scaled up/down by adding/removing VMs	Scaled by adding/removing backend servers
Resource Usage	Consumes CPU, memory, storage	Manages network traffic, minimal resources
Example	Web server, database server	Traffic director for web servers, databases

Q5. (CO4) Explain Cloud Spanner.

Ans- Cloud Spanner is a globally distributed, horizontally scalable, and strongly consistent relational database service provided by Google Cloud Platform (GCP). It is designed to meet the demands of mission-critical, enterprise-grade applications that require high availability, scalability, and ACID (Atomicity, Consistency, Isolation, Durability) transactions across distributed data.

Key features of Cloud Spanner include:

1. Horizontal Scalability: Cloud Spanner scales horizontally across multiple regions and zones, allowing it to handle large volumes of data and high transaction throughput.

2. Strong Consistency: Cloud Spanner offers strong consistency guarantees, ensuring that all reads and writes across the database are immediately consistent, even in the face of network partitions or node failures.

3. Global Distribution: Data in Cloud Spanner can be replicated across multiple regions worldwide, providing low-latency access to users regardless of their geographical location.

4. SQL Interface: Cloud Spanner supports a familiar SQL-based query language, making it easy for developers and DBAs to interact with the database and migrate existing applications to the platform.

5. Automatic Sharding: Cloud Spanner automatically shards data and distributes it across underlying infrastructure, transparently managing partitioning and replication to optimize performance and reliability.

6. High Availability: Cloud Spanner is designed to provide high availability with built-in redundancy and failover mechanisms. It ensures that applications remain accessible and operational even in the event of hardware failures or maintenance activities.

7. Security and Compliance: Cloud Spanner offers robust security features, including encryption at rest and in transit, identity and access management (IAM) controls, and compliance with industry standards and regulations such as HIPAA and GDPR.

Overall, Cloud Spanner enables organizations to build and operate globally distributed, mission-critical applications with high performance, reliability, and scalability while leveraging the familiar SQL interface and cloud-native architecture provided by Google Cloud Platform.

SECTION C (6*2=12 Mark)

Q1. (CO4) List down properties of cloud computing.

Ans- Here are some key properties of cloud computing:

- **On-demand self-service:** Users can provision and configure computing resources without needing to interact with a cloud provider's IT staff.
- **Broad network access:** Cloud services are accessible from anywhere over the internet using a variety of devices.
- **Resource pooling:** The provider's computing resources are pooled to serve multiple users dynamically.
- **Rapid elasticity:** Users can quickly scale resources up or down to meet changing demands.
- **Measured service:** Cloud providers measure resource usage, allowing users to pay only for what they use.
- **High availability:** Cloud providers design their infrastructure for high availability to minimize downtime.
- **Security:** Cloud providers implement security measures to protect user data and applications.
- **Scalability:** Cloud computing allows users to scale their resources up or down to meet changing demands.
- **Pay-per-use:** Users only pay for the resources they use, which can be more cost-effective than traditional IT infrastructure.
- **Multi-tenancy:** Multiple users share the underlying cloud infrastructure, but each user's data and applications are isolated.
- **Resiliency:** Cloud-based systems are designed to be resilient to failures, ensuring high availability of data and applications.

Q2. (CO4) What are the basic services considered in the Storage and Big Data?

Ans- Here are some basic services considered in Storage and Big Data:

Storage Services:

- **Object Storage:** Stores unstructured data like images, videos, audio files, and documents. It's highly scalable and cost-effective for large datasets. (e.g., Google Cloud Storage, Amazon S3)
- **Block Storage:** Provides raw disk volumes for storing data that needs traditional file system structure, often used for running applications on VMs. (e.g., Google Cloud Persistent Disk, Amazon Elastic Block Store)
- **File Storage:** Offers a familiar file system experience accessible from various devices and allows sharing files within an organization. (e.g., Google Drive, Dropbox)
- **Archive Storage:** Provides low-cost, long-term storage for infrequently accessed data. (e.g., Google Cloud Coldline Storage, Amazon Glacier)

Big Data Services:

- **Data Lakes:** Centralized repositories for storing large amounts of structured, semi-structured, and unstructured data. (e.g., Google Cloud Storage, Amazon S3)
- **Data Warehouses:** Optimized for storing and analyzing large datasets for business intelligence purposes. (e.g., Google BigQuery, Amazon Redshift)
- **Data Processing Frameworks:** Tools and libraries for processing and analyzing large datasets, such as Apache Hadoop and Apache Spark. (e.g., Google Cloud Dataproc, Amazon EMR)
- **Big Data Analytics Services:** Managed services for running big data analytics workloads, offering pre-configured environments and tools. (e.g., Google Cloud Dataproc, Amazon EMR)

Additional Considerations:

- **Data Security:** Both storage and big data services should offer robust security features to protect sensitive data.
- **Data Management:** Tools and services for managing data access, lineage, and governance are crucial for large datasets.
- **Scalability:** Storage and big data solutions should be able to scale to accommodate growing data volumes.
- **Integration:** Look for services that integrate well with other tools and applications in your data ecosystem.

SECTION D (10*2=20Mark)

Q1. (CO4) What is persistent Disk and Big Query?

Ans-Persistent Disk and BigQuery are two important services provided by Google Cloud Platform (GCP) for storage and big data analytics, respectively.

1. Persistent Disk:
    - Definition: Persistent Disk is a high-performance block storage service offered by Google Cloud Platform (GCP). It provides durable and reliable storage for virtual machine instances running on Google Compute Engine.
    - Features:
        - High Performance: Persistent Disk offers low-latency, high-throughput storage for virtual machine instances, supporting a variety of disk types optimized for different workloads.
        - Durability: Data stored on Persistent Disk is replicated across multiple physical disks and geographic locations to ensure durability and availability.
        - Scalability: Persistent Disk scales dynamically to meet the storage requirements of virtual machine instances, allowing users to resize disks and add more storage capacity as needed.
        - Snapshots: Persistent Disk supports point-in-time snapshots, enabling users to create backups of their disk data for disaster recovery, data protection, and replication purposes.
2. BigQuery:
    - Definition: BigQuery is a fully managed, serverless data warehouse and analytics platform provided by Google Cloud Platform (GCP). It enables organizations to analyze massive datasets quickly and cost-effectively using SQL queries.
    - Features:
        - Scalability: BigQuery is designed to handle petabytes of data and supports automatic scaling of compute resources based on query demand, allowing users to process large datasets without manual tuning or provisioning.
        - Serverless: BigQuery is serverless, meaning that users do not need to manage infrastructure, perform software updates, or worry about capacity planning. Google handles all aspects of the service, including maintenance and scaling.
        - SQL Interface: BigQuery provides a familiar SQL interface for querying and analyzing data, making it accessible to a wide range of users, including data analysts, data scientists, and business users.
        - Integration: BigQuery integrates seamlessly with other Google Cloud services, such as Google Cloud Storage, Google Data Studio, and Google Sheets, as well as popular third-party tools and frameworks like Tableau, Looker, and Apache Beam.
        - Real-time Analytics: BigQuery supports real-time analytics through streaming ingestion of data, enabling users to analyze and derive insights from data as it arrives in real-time.
        - Security and Compliance: BigQuery offers robust security features, including encryption at rest and in transit, access controls, audit logging, and compliance with industry standards and regulations.

Q2. (CO3) What is Kubernetes in cloud?

Ans- Kubernetes, often shortened to K8s, is an open-source system designed for automating the deployment, scaling, and management of containerized applications in the cloud. Here's a breakdown of Kubernetes in the cloud context:

- **Containerization:** It focuses on containerized applications, which are self-contained software packages including their dependencies (code, libraries, runtime) that can run on any platform with a container engine.
- **Deployment and Management:** Kubernetes automates tasks like deploying containerized applications across a cluster of machines (cloud or on-premises), scaling those applications up or down based on demand, and managing container health.
- **Cloud Agnostic:** While major cloud providers (like Google Cloud Platform, Amazon Web Services, Azure) offer managed Kubernetes services, Kubernetes itself is cloud-agnostic. It can be deployed on various cloud platforms or even on-premises infrastructure.

Benefits of Kubernetes in the Cloud:

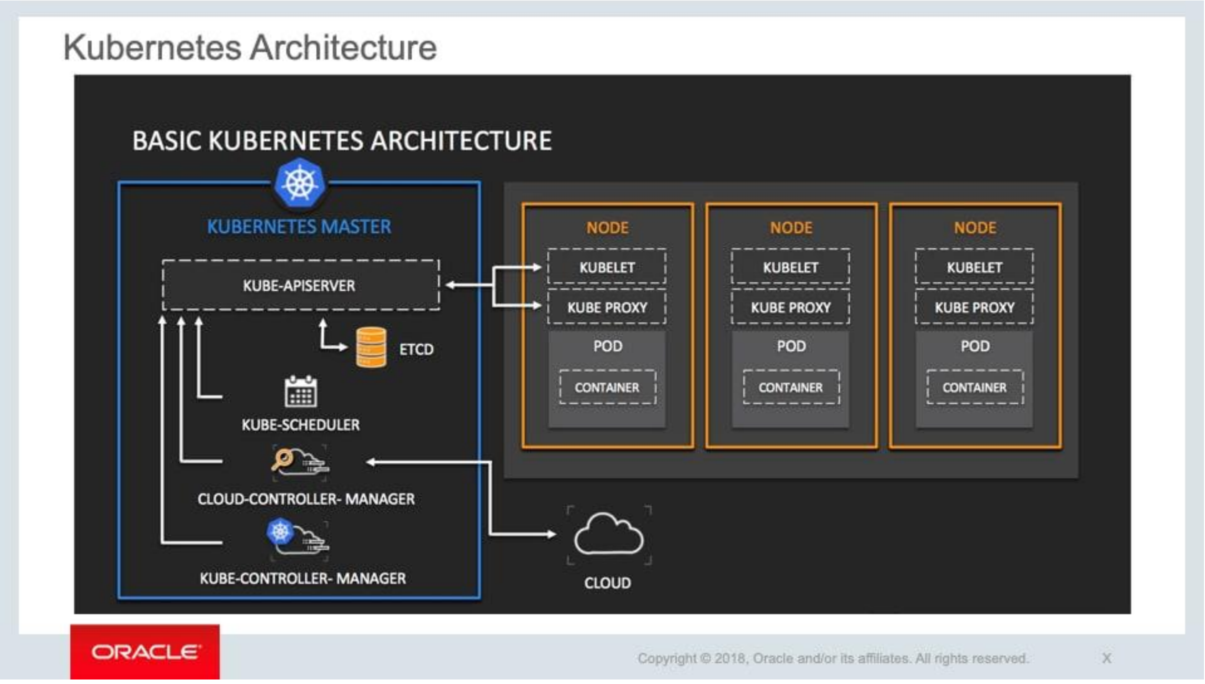
- **Simplified Deployment and Management:** Automates tasks, reducing manual effort and improving efficiency.
- **Scalability:** Easily scales applications up or down to handle changing workloads.
- **High Availability:** Ensures applications stay up and running even if individual machines fail.
- **Resource Optimization:** Optimizes resource utilization by efficiently allocating resources to containers.
- **Portability:** Cloud-agnostic nature allows applications to be easily migrated between cloud platforms.

Cluster

Is a set of machines individually referred to as nodes used to run containerized applications managed by Kubernetes.

What are the components of Kubernetes?

The key components of Kubernetes are clusters, nodes, and the control plane. Clusters contain nodes. Each node comprises a set of at least one worker machine. The nodes host pods that contain elements of the deployed application. The control plane manages nodes and pods in the cluster, often across many computers, for high availability.



The control plane contains the following:

- Kubernetes API server: provides the programming interface (API) for controlling Kubernetes
- etcd: a key-value store for cluster data
- Kubernetes scheduler: matches new pods to available nodes
- Kubernetes-controller-manager: runs a number of processes to manage node failure, control replication, join services and pods via endpoints, and control accounts and access tokens
- Cloud-controller-manager: helps manage APIs from specific cloud providers around such aspects as specific infrastructure routes and load balancing

The node components include:

- kubelet: an agent that checks that containers are running in a pod
- Kubernetes network proxy: maintains network rules
- Docker, containerd, or another type of container runtime